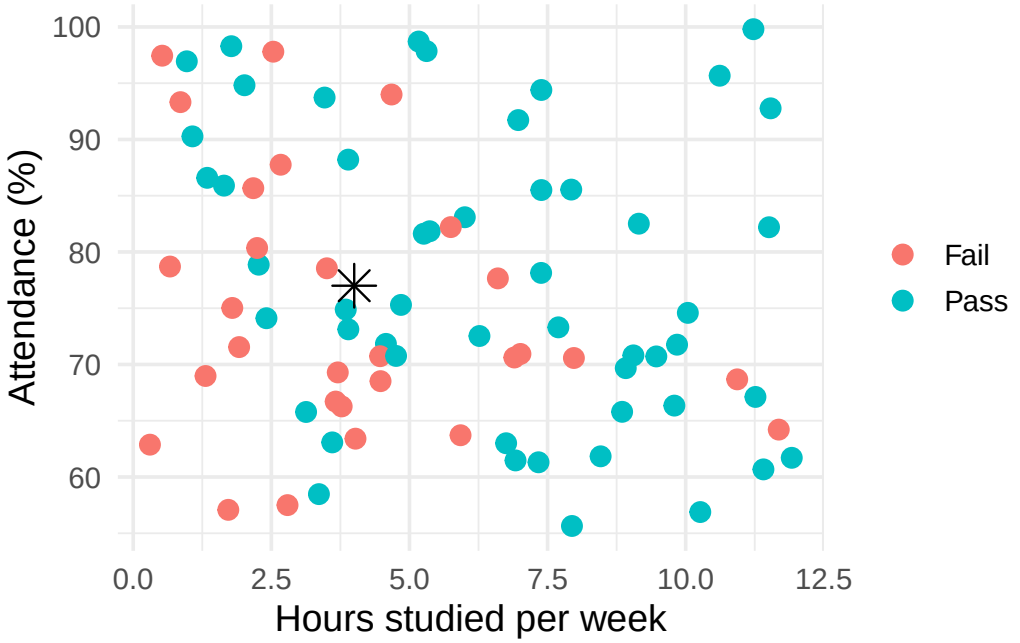
Classification with k-nearest neighbors

Part 1: Make a prediction



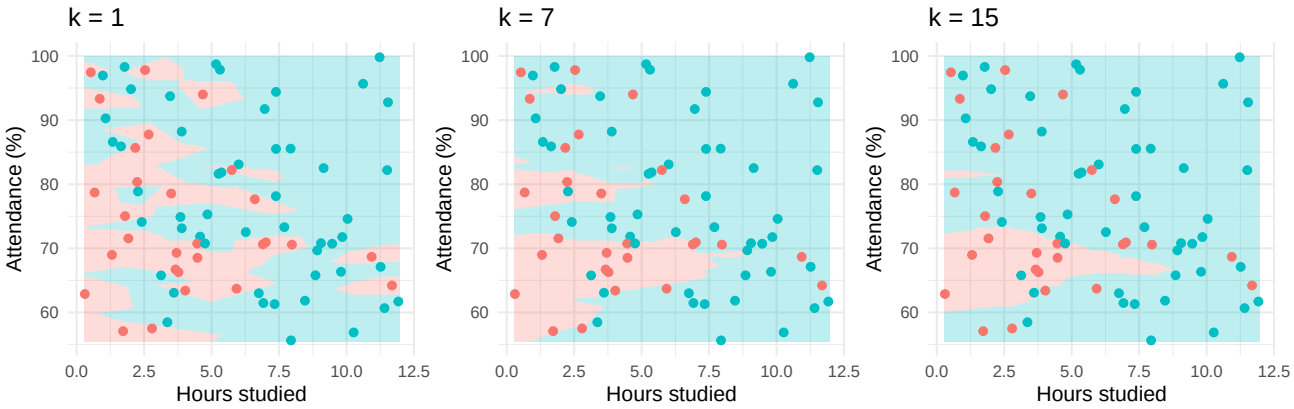
The figure shows reported hours studied per week and attendance rates of 40 students. The points are coloured according to a test result, pass or fail. The star indicates a *new* student.

1.1) What is your guess: will the *new* student pass or fail?

1.2) Circle the student nearest to our new student. Use the nearest student's result to predict the *new* students' result. What is the prediction? $\hat{y} = ?$

1.3) Find the $k = 3$ nearest students to our new student. Use the 3 nearest students' result to predict the *new* students' result. What is the prediction? $\hat{y} = ?$. What is your decision rule (e.g. the mean? the mode?)?

Part 2: Choosing k



The figures show what we would predict for new students at any location in the data domain.

2.1) At what k is the KNN prediction model most flexible? Flexible = highly adapting to the data that we have seen.

2.2) Which value of k do you think generalizes best to new data?

2.3) The data set of 80 students presented above is our *training* data, i.e. the data we use to create a prediction model. For any new student, the predicted grade (pass/fail) is given by the k neighbours in the training data. In KNN classification, our 'prediction model' is based entirely on the training data and our choice of k (a 'hyperparameter').

Now, we also have access to 40 previously unseen students (both hours studied, attendance and grade). These students make up our *validation* data. Error rates on the validation for different choices of k is given below.

Which value of k would you choose? (This is called hyperparameter tuning)

2.4) Now that you have chosen k you have a prediction model: a prediction for a new student is based on it's k nearest neighbors in the training data. Can you say how good your model is at predicting grades of *new* students?

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k = 1	k = 3	k = 5	k = 7	k = 10	k = 15	k = 20
22.5%	20.0%	27.5%	30.0%	32.5%	25.0%	25.0%

Part 3: Training, validation and testing

3.1) Fill inn what each dataset is used for:

Traning data: _____

Validation data: _____

Test data: _____

3.2) What could we do if we did not have a separate validation set?

_____

Part 4: Cross validation

4.1) The table below gives error rates based on 5-fold cross validation on the training data. Which value of k would you choose?

_____

4.2) Is your choice of k the same of different than before? Why do you think it differs?

_____

k = 1	k = 3	k = 5	k = 7	k = 10	k = 15	k = 20
47.5%	38.8%	40.0%	35.0%	32.5%	40.0%	43.8%